



Overview of SurviveX

Software Engineering COMP 4110
University of Massachusetts Lowell
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Team members:

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Project Overview

- “Survive X” is a game that help teaches and sharpen algebra skills for 6th - 8th graders
- This is achieved through a series of tasks the student has to do to survive everyday, and progress to the next island to keep surviving.
- The covered material is all about solving for an unknown variable in an equation
- The student will need to solve math problems to gather resources and survive the zombie infested islands. There are multiple of these islands, which the student will have to go to once resources on the current island run out.

Features

- Use keyboard to explore using top-down 2D movement move around and fight zombies
- The world generation system creates 4 unique islands each game with different biomes and resources to discover
- There is a shop on the main island so the student can buy equipment to help fight zombies
- Weapons and armor are upgradeable through a tier system
- The game will gradually increase difficulty as time passes in the game
- The core feature is the math challenge system which features algebra problems for 6th - 8th grade students

Project Goals

- We wanted to create a game that was equally, academically intensive as fun
- Create a game that could be played during 1 class session which is around an hour
- We also wanted to create a game that is playable by all no matter how sharp your math skills are
- To have students play a game that's easy to pick up but not easy to play
- To give a familiar style game so students are more likely to keep playing it

Domain Research

Our group researched through *The Massachusetts Department of Education*, and looked through the math curriculum to fit standards.

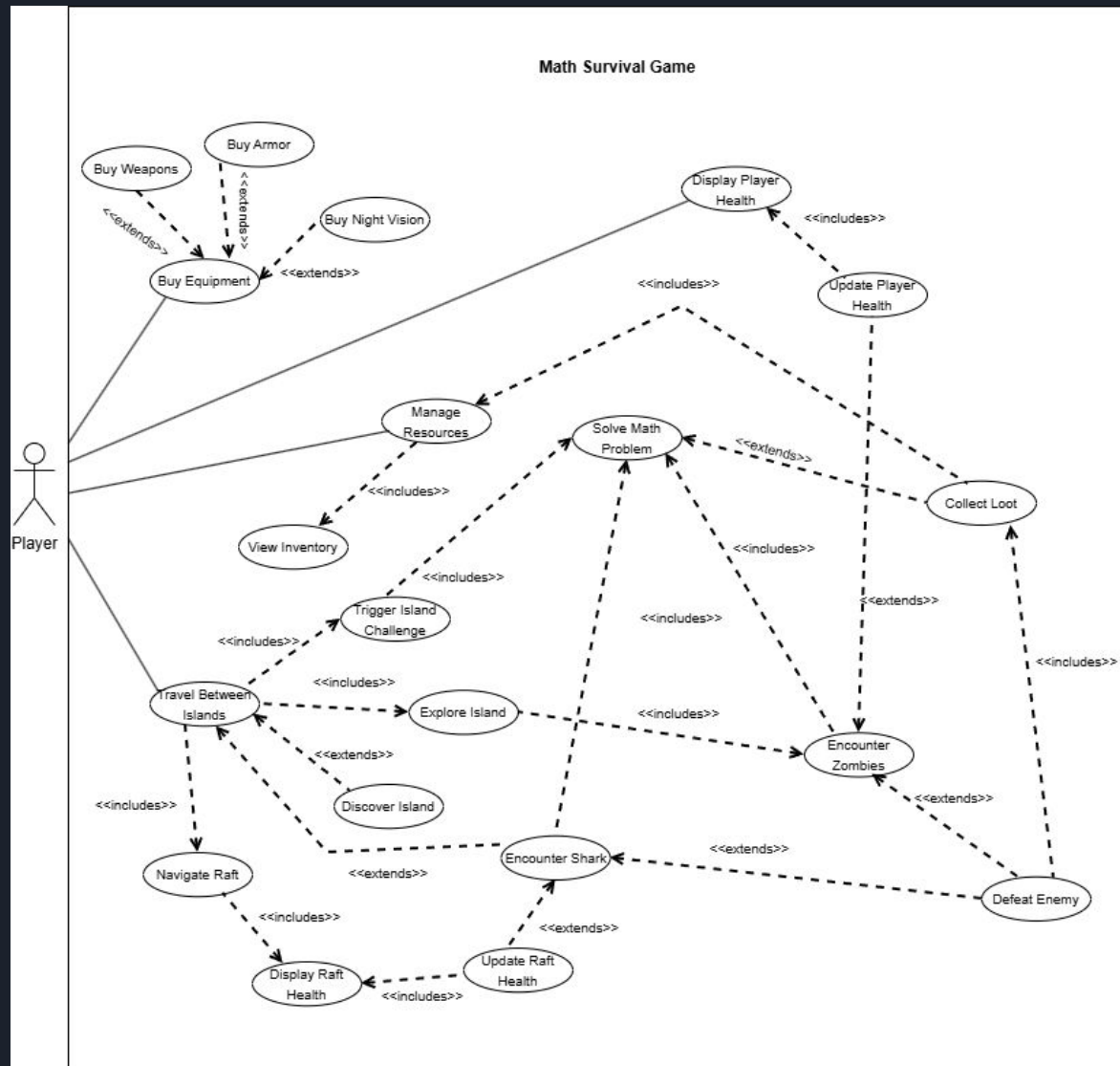
Project Constraints

1. SurviveX must operate on standard classroom and home computers with limited processing
2. The game must use Godot and the Godot game engine in order to create the game
3. The system must also ensure that player and raft health values, enemy damage, inventory updates, and timed challenges behave reliably and cannot desynchronize or freeze, as these are core to the survival loop
4. Math content must remain in line with Massachusetts classroom standards



GODOT
Game engine

Use Case Diagram



Examples





Main Menu

SurviveX

Start Game

Options

Instructions

Exit



Pause Menu

SurviveX

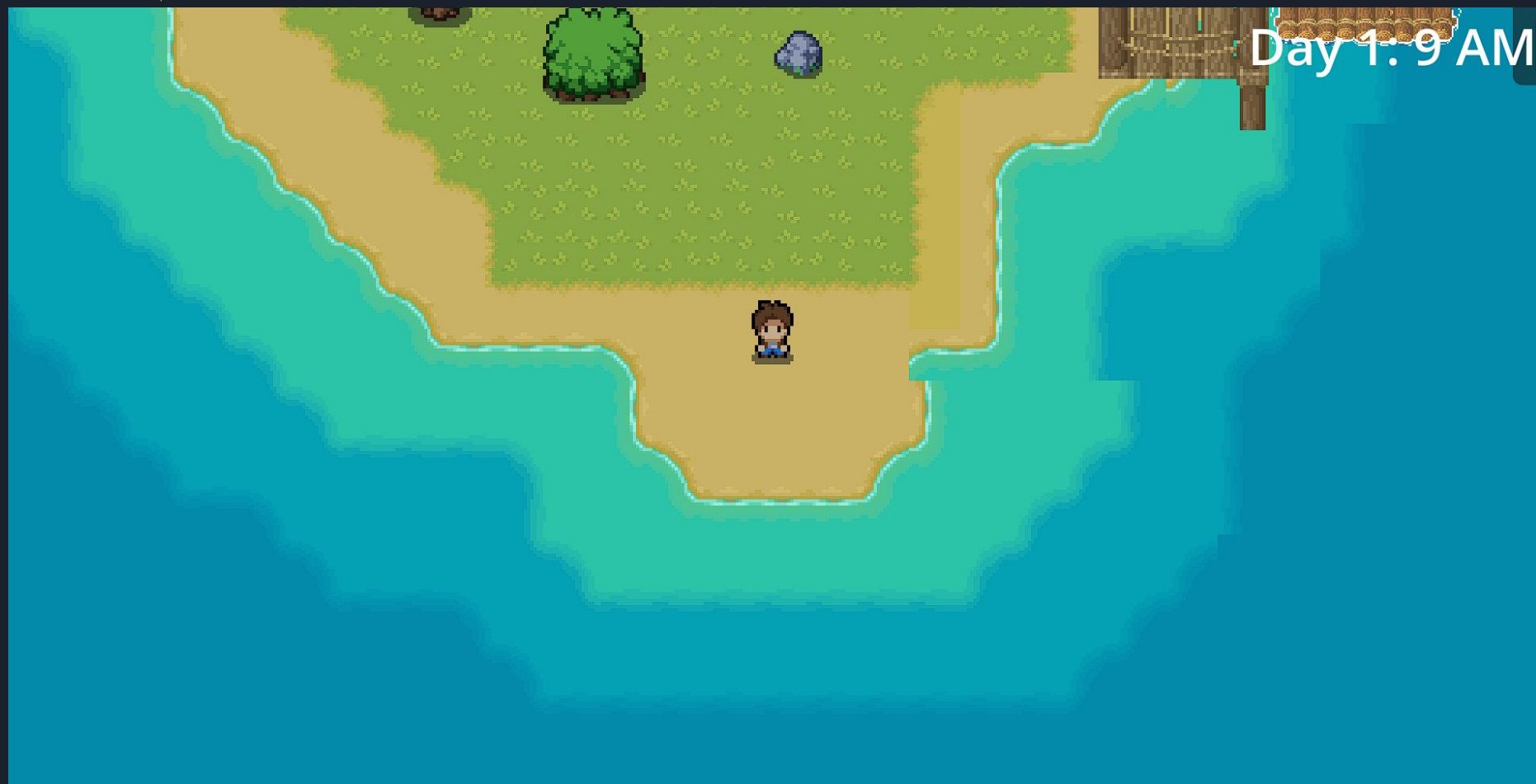
Resume Game

Options

Instructions

Main Menu

Starting Island



Shop Pt. 1



The shop is where the user can purchase armor, weapons, and upgrades which will help the user fight zombies more easily.

Shop Pt. 2



Each item is priced differently, and the shop will display the price of the current item being shown.

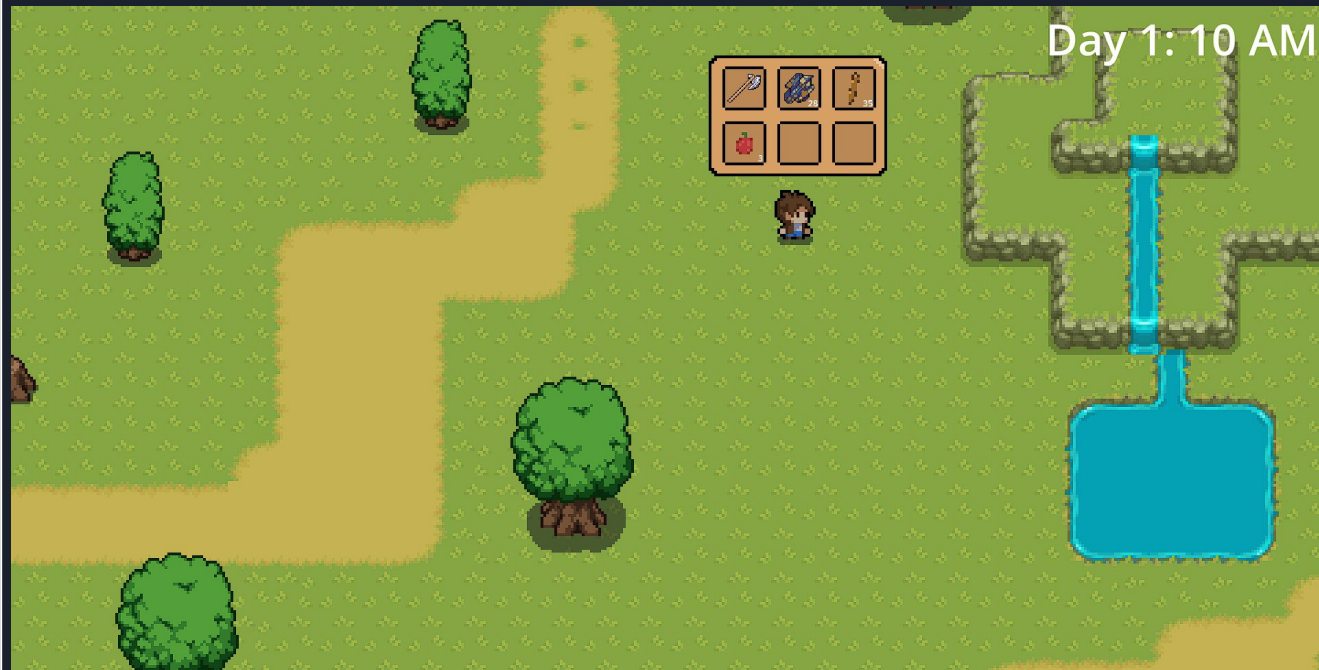
Shop Pt. 3



The shop also allows for easy navigation between items with the two arrows. As well, the buy button will turn green when the user has enough resources to buy the current item.

Inventory

- The player will have a live updating inventory
- The inventory can be toggled on and off
- The inventory keeps track of each item the player has and how much of each item the player has



Using Raft Pt. 1

- In order to travel between islands the player will be able to interact with a raft
- When the player enters the raft, the menu of which island to travel too will show



Using Raft Pt. 2

- When the player clicks on an island to travel too, a math challenge will appear
- The player will need to answer correctly to travel to the selected island



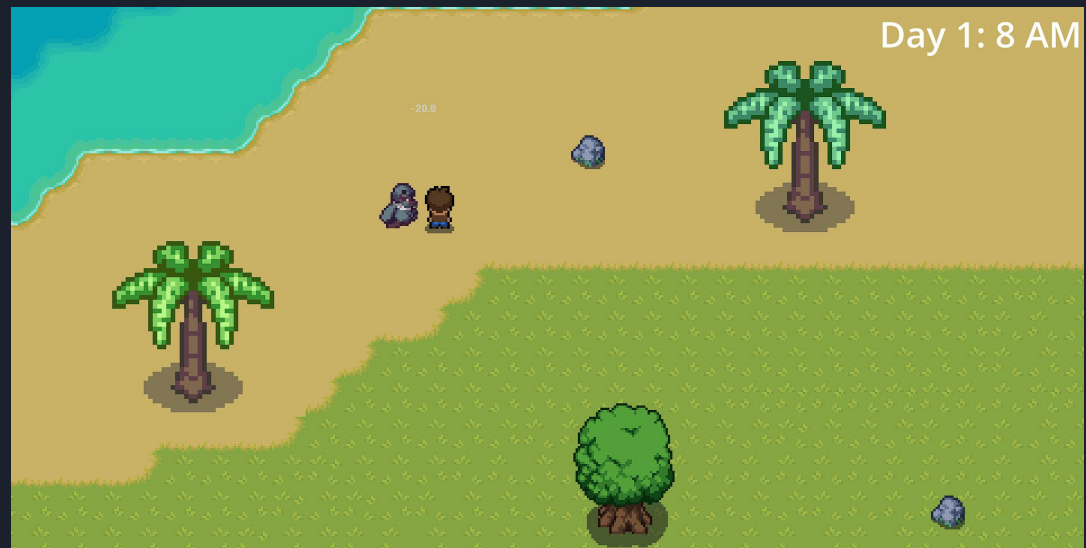
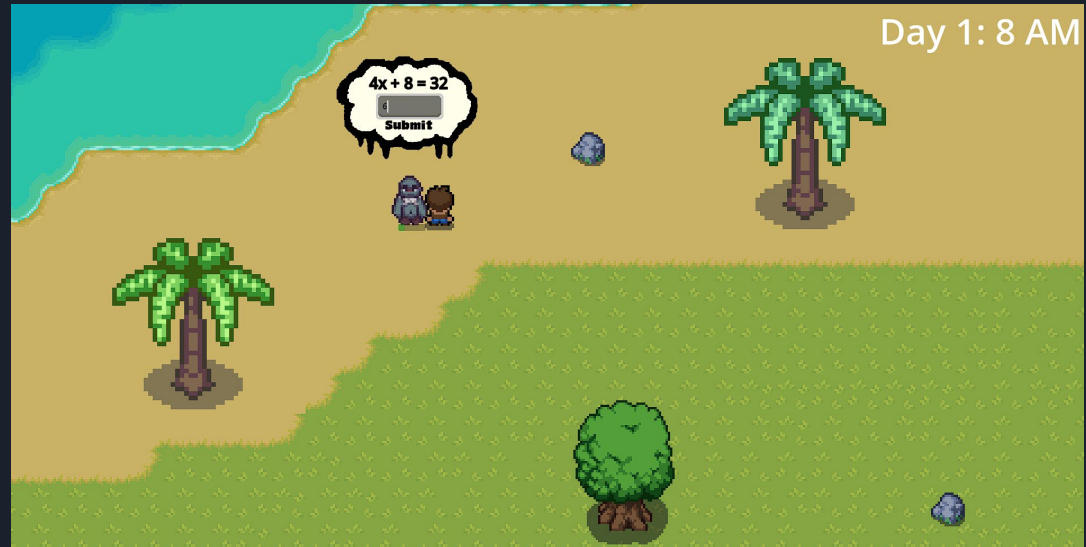
Using Raft Pt. 3

- After answering the question correctly, the player is teleported to the selected island



Fighting Zombie

- Fight zombies by approaching them and solving the question that pops up
- Each question answered correctly does damage to the zombie
- Each question answered incorrectly the user takes damage
- When the zombie or player is not at full health, their health bar is displayed below them



Acknowledgements

We gratefully acknowledge and appreciate the participation of students in 6th - 8th grade, their parents, and their teachers. As well, thank you to anyone who gave feedback on our first version.

Thank You,
Any Questions?

